A Source-Initialized Period-Four Controller in Adaptive Nesterov Iteration

Abstract

We analyze the literal window-one practical adaptive Nesterov rule of Hu, Pollock, Xue and Zhu on a scalar positive-definite quadratic. An exact interval certificate proves an attracting period-four normalized controller, selected by the printed initialization, while the primal iterates converge geometrically. The result holds for every fixed curvature in $[0.7499, 0.7501]$. At curvature $3/4$ we give sharper cycle and rate enclosures. A coexisting stable normalized fixed point, source indexing qualifications, and the limits of the comparison with exact-spectral-radius theory are included.

Keywords: adaptive Nesterov iteration; period-four controller; source initialization; interval arithmetic; geometric convergence

2020 Mathematics Subject Classification. Primary 37N40; Secondary 39A23, 65G30, 65K10.

1 Literal algorithm and normalized state

For $f(x) = \lambda x^2/2$, $0 < \lambda < 1$, set $q = 1 - \lambda$ and take any $x_0 \neq 0$. We specialize Algorithm 2.2 and equation (2.5) of arXiv:2602.13620v1 [1] to a unit step and a window of length one:

$$\begin{aligned} &x_1 = qx_0, \qquad \rho_1 = |r_1|/|r_0| = q, \\ &\beta_k = \frac{\rho_{k-1}}{2 - \rho_{k-1}}, \qquad x_k = q\big((1+\beta_k)x_{k-1} - \beta_k x_{k-2}\big), \quad k \geq 2, \\ &\rho_k = \frac{\|R_k\|_2}{\|R_{k-1}\|_2} = \sqrt{\frac{x_k^2 + x_{k-1}^2}{x_{k-1}^2 + x_{k-2}^2}}, \qquad R_k = (r_k, r_{k-1}), \quad r_k = -\lambda x_k. \end{aligned} \tag{1}$$

The initial estimate is ordinary and exceptional; later estimates are stacked. No clipping or restart is added. In particular no artificial r_{-1} is needed.

Where ratios are nonzero, put $t_k = x_k/x_{k-1}$, $z_k = (t_k, \rho_k)$. Direct substitution gives the exact map

$$F_q(t, \rho) = (u, v), \quad u = q\frac{2 - \rho/t}{2 - \rho}, \qquad v = |t|\sqrt{\frac{1+u^2}{1+t^2}}, \qquad z_1 = (q, q). \tag{2}$$

The symbol u in (2) is the first output coordinate. All nonzero and positivity conditions needed to justify this chart are proved below. As a check at $q = 1/4$, $\beta_2 = 1/7$, $x_2 = x_0/28$, $t_2 = 1/7$, and $\rho_2 = \sqrt{50/833}$.

2 Statements

Theorem 1. *For each* fixed $\lambda \in [7499/10000, 7501/10000]$*, recurrence* (1) *is well-defined for all time, with* $x_k \neq 0$*,* $|t_k| > 0.04$ *and* $0 < \rho_k < 0.34$*. Its normalized startup converges exponentially by phase to an attracting exact-period-four orbit of* (2)*. The four limiting* ρ *coordinates are distinct; hence* ρ_k *and* β_k *do not converge. There is a number* r_λ *with*

$$0.1601 < r_\lambda < 0.1604, \qquad \lim_{k\to\infty} |x_k/x_0|^{1/k} = r_\lambda.$$

In particular the primal orbit converges to its minimizer. The statement is uniform over fixed curvatures and asserts nothing about arbitrarily time-varying curvature.

Theorem 2. *At* $\lambda = 3/4$*, the four normalized phases, in the order starting at the limit of* z_{401+4j}*, have the following approximate coordinates (only for orientation):*

$$\begin{aligned}
&(0.3356887440591454,\ 0.1307469335035425),\\
&(0.2153950480433309,\ 0.3255353223252715),\\
&(0.07295752716358487,\ 0.2111254787545297),\\
&(-0.1249128310826865,\ 0.07332960959851113).
\end{aligned}$$

Their signed first-coordinate product P *and the root rate satisfy the rigorous strict enclosures*

$$\begin{aligned}
-0.000658945734189712520431 &< P\\
&< -0.000658945734189712519037,\\
0.160218408231717313490625 &< |P|^{1/4}\\
&< 0.160218408231717313575260.
\end{aligned}$$

For each residue $j = 0, 1, 2, 3$*,* $x_{401+4n+j}/P^n$ *tends to a finite nonzero constant. The one-step ratios, including their absolute values, have distinct phase limits and do not converge.*

3 Validated contraction framework

All decimal constants in certificates represent exact rationals. We give the continuous-region proof and the finite algorithms underlying the numerical bounds, so that trajectory plots or unverified numerical roots are not premises. For u, v in (2), differentiation on a nonzero-sign t region gives

$$\begin{aligned}
u_t &= \frac{q\rho}{(2-\rho)t^2}, & u_\rho &= \frac{q(2-2/t)}{(2-\rho)^2}, & u_q &= \frac{2-\rho/t}{2-\rho},\\
v_t &= v\left(\frac{1}{t(1+t^2)} + \frac{uu_t}{1+u^2}\right), & v_\rho &= \frac{vuu_\rho}{1+u^2}, & v_q &= \frac{vuu_q}{1+u^2}.
\end{aligned} \tag{3}$$

These formulas include the dependence of v on u and the absolute-value branch. For $G_q = F_q^4$, state and parameter derivatives are propagated through four steps by

$$J_{j+1} = A_j J_j, \qquad b_{j+1} = A_j b_j + d_j, \qquad J_0 = I, \quad b_0 = 0,$$

where $A_j = D_z F_q$ and $d_j = \partial_q F_q$ at the corresponding phase.

Lemma 1. *Let $Q = [q_m - h, q_m + h]$, $B = c + P_0[-R, R]^2$, with P_0 nonsingular. Suppose all four phase maps are defined on B and Q, and*

$$\sup_{B\times Q} \|P_0^{-1} D_z G_q P_0\|_\infty \le L < 1,$$
$$\sup_{B\times Q} \|P_0^{-1} \partial_q G_q\|_\infty \le M, \quad \|P_0^{-1}(G_{q_m}(c) - c)\|_\infty \le \delta,$$
$$\delta + Mh + LR < R.$$

Then for each fixed $q \in Q$, G_q has a unique fixed point in B, and every G_q orbit starting in B converges to it exponentially. If the four phase projections are pairwise disjoint, its period under F_q is exactly four.

Proof. Use the norm $\|z\|_{P_0} = \|P_0^{-1} z\|_\infty$. The mean-value inequality on the convex set B and the parameter segment gives $\|G_q(z) - c\|_{P_0} \le L\|z - c\|_{P_0} + Mh + \delta < R$. Thus G_q is a contraction of the complete closed set B into itself. Banach's theorem gives the unique fixed point and exponential attraction. A smaller period would identify two distinct phase projections, a contradiction. ∎

4 Proof of the uniform interval theorem

Proof of Theorem 1. Partition $q \in [2499/10000, 2501/10000]$ into exactly 10000 adjacent closed cells of width $2 \cdot 10^{-8}$. Use

$$P_0 = \begin{pmatrix} 51/100 & -12/100 \\ -85/100 & 0 \end{pmatrix}, \qquad R = 10^{-5}.$$

Its determinant is $-102/1000$. For each midpoint q_m, form a rational candidate center by starting at (q_m, q_m) and taking 50 blocks of four interval-enclosed steps, rounding each center to a rational midpoint. This merely proposes a center; it is not assumed to be a true root.

The portable certificate `uniform.py` evaluates (3) on the rectangle hull of each parallelogram $B = c + P_0[-R, R]^2$. This larger hull preserves inclusion. It verifies all the inequalities of Lemma 1, and pairwise separation of both the t and ρ phase projections, on every cell. Every intermediate input and output t has constant nonzero sign, $0 < \rho < 1$, and every denominator and square-root argument is valid on its full box.

For completeness, startup enclosures retain parameter dependence. The initial center is (q_m, q_m) and the radius encloses $P_0^{-1}([-h,h]^2)$. If a block state lies in $z + P_0[-r,r]^2$, then

$$\|G_q(z_{\text{actual}}) - G_{q_m}(z)\|_{P_0} \le L_{\text{block}} r + M_{\text{block}} h.$$

The next center is the rational midpoint enclosure of $G_{q_m}(z)$; its rounding radius is added to the displayed bound. Thus induction supplies an enclosure of every prescribed startup throughout each parameter cell. The comparison $\|z - c\|_{P_0} + r < R$ certifies entry, in at most 100 updates. A separately implemented full-coverage calculation gives the stronger maximum of 92 updates and the convenient rational summaries

$$L < 0.905, \qquad |t| > 0.04056, \qquad \rho < 0.33795, \qquad 0.160119 < r_\lambda < 0.160318.$$

The broader bounds in the statement avoid reliance on displayed rounded extrema. All 10000 cells, including endpoints, are checked; cell width is not the total interval width. No continuity argument replaces parameter validation.

Lemma 1 proves cycle existence and attraction for every fixed parameter. Endpoint conclusions from adjacent cells are compatible because both attract the same prescribed startup. The infinite tail is covered by invariance and its four phase images; the prefix domain bounds and tail domain bounds together prove $x_k \neq 0$ and well-definedness of (1). Reconstruction uses $x_k = t_k x_{k-1}$, $x_1 = q x_0$, and exactly recovers the stacked estimate.

Let P_q be the signed product of the four limiting t coordinates. The certificate multiplies the four phase intervals and verifies $P_q < 0$, then obtains $|P_q|^{1/4}$ by two outward square roots. Phase convergence is exponential; the maps have bounded derivatives on their certified tubes and $|t|$ stays away from zero. Hence deviations of $\log|t_k|$ from the corresponding phase values are absolutely summable. Averaging $\log|x_k/x_0| = \sum_{j=1}^{k} \log|t_j|$ proves the root rate. Pairwise distinct ρ phases prove its nonconvergence; strict monotonicity of $\rho/(2-\rho)$ gives the momentum conclusion, with an index shift of one. ∎

5 Sharper certificate at curvature three quarters

Proof of Theorem 2. The independent exact-parameter program `exact_quarter.py` uses a lattice of spacing 10^{-400}. Let c be its explicitly specified rational center and $X = c + [-10^{-20}, 10^{-20}]^2$. Set $H = F_{1/4}^4 - I$. A rational preconditioner C is obtained from the rounded midpoint inverse of $DH(c)$, and its determinant is checked nonzero. Full-box calculation gives

$$\sup_X \|I - CDH\|_\infty < 1.93 \cdot 10^{-17}.$$

The enclosure of $c - CH(c) + (I - CDH(X))(X - c)$ lies strictly inside X; its displacement divided by 10^{-20} is less than $1.93 \cdot 10^{-17}$. Therefore $z \mapsto z - CH(z)$ contracts X into itself. Its unique fixed point satisfies $H = 0$ because C is nonsingular. Pairwise separation of all four phase projections proves exact period four.

Existence alone does not prove initialized attraction. On the larger box $B = c + [-10^{-12}, 10^{-12}]^2$, exact interval differentiation proves

$$\sup_B \|DF_{1/4}^{20}\|_\infty < 0.104334 < 0.11, \quad \|F_{1/4}^{20}(c) - c\|_\infty < 2 \cdot 10^{-70}.$$

Their sum after multiplication by the radius is less than the radius. Thus F^{20} contracts B into itself, and its unique fixed point is the already certified period-four point in X. The direct rational startup from $z_1 = (1/4, 1/4)$ reaches $z_{401} = F^{400}(z_1)$ strictly inside B, with

$$.335688744059145422701936 < t_{401} < .335688744059145422701937,$$
$$.130746933503542450833406 < \rho_{401} < .130746933503542450833407.$$

The index is 401, not 400. All finite startup boxes satisfy $|t| > .0406002593$, $0 < \rho < .3373728550$; all twenty tail boxes satisfy $.0729575167 < |t| < .3356887502$, $0 < \rho < .3255353292$. Thus the full recurrence and derivative branches remain valid forever.

Contraction gives geometric phase convergence with block factor 0.11; continuity with bounded derivatives covers intermediate phases. Multiplication of the four certified phase intervals gives the product and root-rate bounds in the statement. If A_n is a successive four-step physical multiplier in any fixed phase, then $A_n/P = 1 + e_n$ with $\sum |e_n| < \infty$. Since none of these multipliers vanish, the corresponding product converges to a finite nonzero constant after any finite initial segment. This proves the refined residue-class limit and the eventual sign reversal every four steps. ∎

6 Coexistence in normalized state space

Proposition 6.1. For every $q \in (0, 1)$ the normalized map has a locally asymptotically stable fixed point (r, r), where $r = 1 - \sqrt{1 - q}$.

Proof. Since $q = r(2 - r)$, direct substitution gives the fixed point. Its derivative is

$$J = \begin{pmatrix} 1 & b \\ 1 & br^2/(1 + r^2) \end{pmatrix}, \qquad b = \frac{2(r - 1)}{2 - r}, \qquad D = \det J = \frac{2(1 - r)}{(2 - r)(1 + r^2)}.$$

The three real two-dimensional Schur margins are

$$1 - D = \frac{r(1 + 2r - r^2)}{(2 - r)(1 + r^2)} > 0,$$
$$1 - \operatorname{tr} J + D = -b > 0,$$
$$1 + \operatorname{tr} J + D = 2 + b\frac{r^2 - 1}{1 + r^2} > 2.$$

Thus all eigenvalues lie strictly in the unit disk. Smoothness gives local nonlinear attraction by contraction in an equivalent norm. ∎

Coexistence concerns two-dimensional normalized states. It does not mean varying the nonzero magnitude or sign of x_0 under the printed scalar initialization selects different attractors: all such initializations give the same (q, q) and select the cycle in Theorem 1.

7 Source interpretation and limits

The source [1] permits positive-definite matrices with $L \leq 1$. Our scalar example has true $L = 3/4$; exact rescaling to scalar $L = 1$ would solve it in the first unit step. Its Lemma 2.7 and Theorem 2.8 use exact spectral radii, while practical estimates are treated separately. The present feedback cycle therefore does not contradict that conditional theory. Instantaneous matrix spectral radii alone also cannot be substituted for an analysis of arbitrary nonnormal matrix products.

Section 3.1 of [1] already observes estimator oscillations numerically, interpreted there as ultimately convergent. Persistent period-four feedback with the certified startup refutes a universal interpretation of eventual practical-estimator convergence for this literal infinite recurrence, not the separate exact-spectral-radius theorem. A specific prose indexing discrepancy is resolved by the algorithm: if M_k uses α_k, β_k, then $R_k = M_k R_{k-1}$, equivalently $R_{k+1} = M_{k+1} R_k$, rather than $R_{k+1} = M_k R_k$.

Austin, Pollock and Zhu [2] already discuss fluctuating residual estimates for accelerated power iteration; their clipping and inversion rules differ from (1). O'Donoghue and Candès [3] analyze damped quadratic oscillations and restart, rather than this stacked-residual rational feedback. These are relevant antecedents, so merely observing oscillation is not a novelty claim. The contribution here is the specified periodic controller, its initialized basin, coexistence, and exact parameter certificate. No claim is made about tolerance-stopped runs, floating-point limiting behavior, arbitrary varying λ, a maximal cycle interval, practical divergence, or a nonminimizing primal cycle.

A Portable interval certificate

Run `python uniform.py --workers 4` and `python exact_quarter.py` in the accompanying certificate directory. The former uses Python integers, `Fraction`, and `isqrt`, with lattice spacing 10^{-80}; worker count changes scheduling only. The latter uses the same standard-library primitives at spacing 10^{-400}. The files contain exact centers and all executable assertions. A separate `uniform_independent.py` is included as a second full-coverage implementation.

Products are bounded by extrema of the four endpoint products; divisions reverse positive reciprocal endpoints and check domains. For lattice denominator D, the lower square-root integer is $\mathrm{isqrt}(\lfloor qD^2 \rfloor)$ and one additional lattice unit bounds above. These operations enclose every map and derivative value inductively. Signed Jacobian products are formed before norm bounds, preserving cancellation while retaining interval inclusion. Rounding the inverse of P_0 outward can only enlarge bounds. These finite exact computations, together with Lemma 1 and the tail proof, constitute the certificate; no numerical root solver or floating-point trajectory is assumed correct.

References

[1] Hu, Pollock, Xue and Zhu. An adaptive framework for first-order gradient methods. arXiv:2602.13620v1, 2026. https://arxiv.org/abs/2602.13620v1.

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